

Novelty Assessment and Next Steps

For the Proof of Conjecture 3: The Period Length at $\omega = 1$

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Abstract

We assess the novelty of the theorem $\lambda_{\alpha,\beta} = \alpha + \beta + 1$ for one-dimensional cut-and-project sequences at window width $\omega = 1$. An extensive survey of related literature — including the three-distance theorem, Christoffel word theory, and the cut-and-project literature on model sets — reveals no prior result that states or proves this formula. We conclude that the theorem is almost certainly novel, identify the closest related known results, and provide recommendations for publication.

1 The Theorem Under Review

The result whose novelty is assessed here is the following.

Theorem 1. *Let $\alpha, \beta \in \mathbb{N}$ with $\gcd(\alpha, \beta) = 1$, and let $\omega = 1$. Consider the lattice points $(x, y) \in \mathbb{Z}^2$ satisfying*

$$\frac{\alpha}{\beta}(x-1) \leq y \leq \frac{\alpha}{\beta}x + 1,$$

projected vertically onto $f(x) = \frac{\alpha}{\beta}x$. The period length of the resulting distance sequence $(\delta^{(i)})$ equals

$$\lambda_{\alpha,\beta} = \Lambda_{\alpha,\beta} := \alpha + \beta + 1.$$

The proof (see [1]) proceeds in four steps:

1. The invariant $r := \alpha x - \beta y$ takes values in $\mathcal{R} = \{-\beta, \dots, 0, \dots, \alpha\}$, giving $|\mathcal{R}| = \alpha + \beta + 1$ distinct values.
2. For each fixed $r \in \mathcal{R}$, the projected values $\tilde{p} = \beta x + \alpha y$ form an arithmetic progression with common difference $\alpha^2 + \beta^2$.
3. By Bézout's identity, the $\alpha + \beta + 1$ residues modulo $\alpha^2 + \beta^2$ are pairwise distinct.
4. The sorted union of these progressions therefore has period exactly $\alpha + \beta + 1$.

2 Related Literature

2.1 The Three-Distance Theorem

The three-gap theorem (Steinhaus, Sós, Surányi, Świerczkowski; 1950s) states that placing n points $\{k\alpha\}$ on a unit circle produces at most three distinct gap lengths [2]. When $\alpha = p/q$ is rational the number of distinct gap lengths reduces to at most two. More recent treatments include the lattice proof by Marklof and Strömbergsson (2017) and the exposition by Berthé and Reutenauer [3].

This theorem is related in spirit but does not imply the result above for two reasons:

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- It concerns rotations on a circle, not lattice projections through a strip of finite width.
- It characterises the *number* of distinct gap lengths (at most three), not the *period length* of the gap sequence. The quantity $\alpha + \beta + 1$ does not appear in this framework.

2.2 Christoffel Words and Sturmian Sequences

The closest structural relative found in the literature is the theory of Christoffel words [4]. The lower Christoffel word of slope p/q with $\gcd(p, q) = 1$ is a word of length $p + q$ over a two-letter alphabet, and the mechanical sequence of rational slope p/q is periodic with period $\mathbf{p} + \mathbf{q}$.

The difference from Theorem 1 is mathematically significant:

- Christoffel words have period $p + q$; the theorem gives period $\alpha + \beta + 1$. The extra $+1$ is a direct consequence of the symmetric window of width $\omega = 1$, which allows r to range over $\alpha + \beta + 1$ values instead of $\alpha + \beta$.
- The Christoffel/Sturmian framework describes a *single* line (window of vanishing width), not a strip of finite width $\omega = 1$ capturing lattice points on both sides of the central line.
- The object of study here is the sequence of *Euclidean distances* between projected points, not a symbolic word period.

2.3 Cut-and-Project Literature (Model Sets)

The mainstream literature on model sets and quasicrystals [5, 6] focuses almost exclusively on irrational slopes, since rational slopes produce trivially periodic sequences. Key topics in this literature — linear repetitivity, patch frequencies, diffraction spectra — differ from the question addressed here. No paper was found that determines the period of the distance sequence for a one-dimensional cut-and-project set with rational slope and a symmetric window of finite width.

2.4 Beatty Sequences and Fraenkel’s Conjecture

The Beatty sequence and Fraenkel conjecture literature concerns partitions of $\mathbb{N}$ into Beatty sequences with rational or irrational parameters. None of the papers in this area address projection distances with a symmetric window of the kind appearing in Theorem 1.

3 Assessment of Novelty

Based on the survey above, the theorem is **almost certainly novel**. The specific reasons are:

1. **The formula $\alpha + \beta + 1$ does not appear in the existing literature** in this context. The nearest analogue — the period $p + q$ of Christoffel words — differs by exactly 1, and this difference is mathematically consequential.
2. **The construction is not a standard object of study.** Most cut-and-project work concerns irrational slopes or asymptotic behaviour. The rational slope case with a symmetric window of width $\omega = 1$ is not treated in the mainstream literature.
3. **The proof strategy is original.** Classifying the $\alpha + \beta + 1$ arithmetic progressions in the projected coordinate $\tilde{p} = \beta x + \alpha y$ via the invariant $r = \alpha x - \beta y$, and then invoking Bézout’s identity to establish distinct residues modulo $\alpha^2 + \beta^2$, does not appear to have been written down before.

Remark 1. A full MathSciNet/zbMATH/Scopus search by a librarian with database access would be a prudent final step before submission. Web-based searches cannot access all specialised journal content.

4 Recommendations for Publication

Recommendation 1. Add a paragraph to the introduction of [1] explaining why Theorem 1 is *not* a direct consequence of Christoffel word theory. The key points to make are:

- The period of a Christoffel word of slope α/β is $\alpha + \beta$, not $\alpha + \beta + 1$.
- The extra $+1$ arises from the symmetric window $\omega = 1$, which widens the strip and adds one additional residue class.
- The result concerns Euclidean distances in the projected sequence, not a symbolic word over a finite alphabet.

Recommendation 2. Cite the following works as the closest known results, to help a reviewer see the gap between the existing literature and the new theorem:

- Berth   and Reutenauer [3] on the three-distance theorem.
- Berstel, Lauve, Reutenauer, and Saliola [4] on Christoffel words.

Recommendation 3. Request a zbMATH/MathSciNet search using the terms “cut-and-project”, “period length”, “distance sequence”, “rational slope”, and “coprime” before final submission, to rule out any overlooked specialised result.

References

- [1] Dirk Kunert, 2025, *Proof of Conjecture 3: The Period Length at $\omega = 1$* , available at: <https://github.com/dkunert/cut-and-project>.
- [2] *Three-gap theorem*, Wikipedia, available at: https://en.wikipedia.org/wiki/Three-gap_theorem, accessed April 2025.
- [3] Val  rie Berth   and Christophe Reutenauer, *On the Three Distance Theorem*, *The Mathematical Intelligencer*, 2024. Available at: <https://link.springer.com/article/10.1007/s00283-023-10316-z>.
- [4] Jean Berstel, Aaron Lauve, Christophe Reutenauer, and Franco V. Saliola, *Combinatorics on Words: Christoffel Words and Repetitions in Words*, CRM Monograph Series, Vol. 27, American Mathematical Society, 2008.
- [5] Marjorie Senechal, *Quasicrystals and Geometry*, Cambridge University Press, 2009. ISBN: 978-0-521-57541-6.
- [6] Alan Haynes, Henna Koivusalo, James Walton, and Lorenzo Sadun, *Gaps Problems and Frequencies of Patches in Cut and Project Sets*, *Mathematical Proceedings of the Cambridge Philosophical Society*, 2016. Available at: <https://www.cambridge.org/core/journals/mathematical-proceedings-of-the-cambridge-philosophical-society/article/abs/gaps-problems-and-frequencies-of-patches-in-cut-and-project-sets/1A47C2BE5E660C5C0A0AC4A828F3B6E7>.